

Applications of Discrete Mathematics in Modelling Computation

Discrete Mathematics plays an important role in computer science because computers work using discrete steps, symbols, and logic. Many computational models are built using concepts from discrete maths. These models help us understand how languages are processed, how machines behave, and how problems are solved.

Language and Grammars – Introduction:

A language in computer science is a set of strings formed using symbols.

A grammar is a set of rules used to generate these strings.

Grammars help us understand how programming languages, compilers, and regex patterns are formed.

Phrase-Structured Grammars & Types:

Grammars are divided into four types called Chomsky Hierarchy:

1. Type 0 – Unrestricted Grammar
2. Type 1 – Context-Sensitive Grammar
3. Type 2 – Context-Free Grammar (CFG)
4. Type 3 – Regular Grammar

Regular grammar is the simplest and is used to build finite automata.

Derivation Trees:

A derivation tree (parse tree) is a tree structure that shows how a string is derived from a grammar rule. It helps in syntax analysis in compilers.

Finite State Machines with Output – Introduction:

These are machines that read input symbols and also produce output.

Types:

1. Moore Machine – Output depends only on the current state.
 2. Mealy Machine – Output depends on current state + current input.
- These are used in vending machines, traffic signals, and digital circuits.

Finite State Machines without Output – Introduction:

These machines don't produce output. They only accept or reject strings. They are mainly used in language recognition.

Set of Strings

A finite automaton accepts a specific set of strings known as a language.

Finite State Automata (FSA)

An FSA is a model with a finite number of states. It processes input one symbol at a time.

Types include:

DFA (Deterministic Finite Automaton)

NFA (Non-Deterministic Finite Automaton)

DFA has exactly one transition per symbol,
NFA can have many transitions for the same symbol.

Language Recognition by FSM:

A Finite State Machine recognizes a language by checking whether the input string ends in an accept state.

If it ends in an accept state → the string is valid.

If not → string is rejected.

This is used in:

Lexical analysis of compilers

Pattern matching

Token generation

Language Recognition – Introduction:

Language recognition is the process of determining whether a string belongs to a particular language.

FSMs, grammars, and automata models are used to recognize and classify languages.

Turing Machine – Introduction & Definition:

A Turing Machine is the most powerful computational model.

It consists of:

An infinite tape

A tape head

A set of states

A transition function

It can perform any computation that a modern computer can do (given unlimited time and memory).

Turing Machines help define which problems are computable and which are not.

Conclusion:

Discrete Mathematics provides the foundation for modelling computation. Languages, grammars, finite automata, and Turing machines form the core of how computers understand input, process data, and decide results. These models are essential in designing compilers, interpreters, digital circuits, and many real-world computing systems.